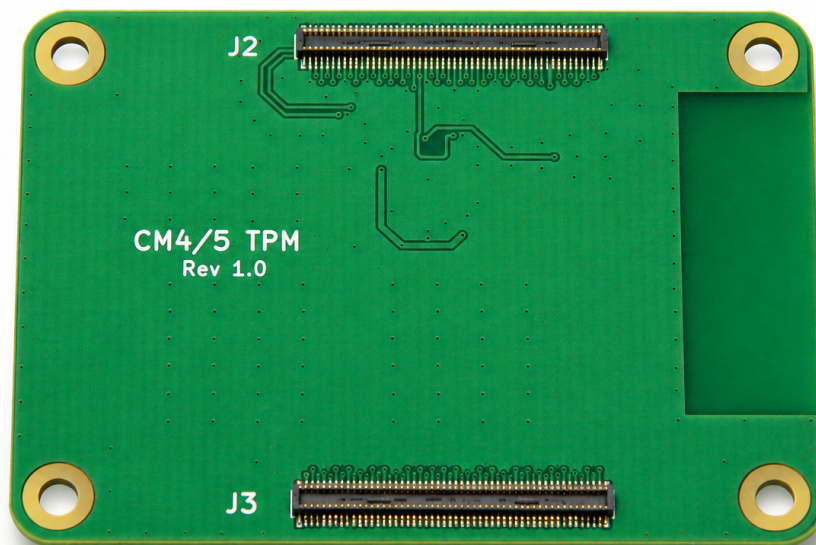
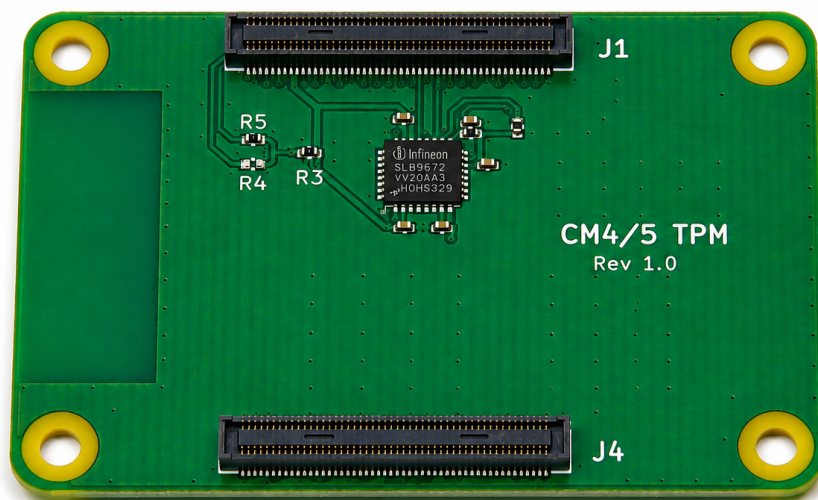


CM4/5 TPM Module

Trusted Platform Module Stack-up for Raspberry Pi Compute Module 4 & 5

Revision 1.0 | September 12, 2026



TPM IC: Infineon OPTIGA™ TPM SLB 9670VQ2.0 **Interface:** SPI **Standard:** TCG TPM 2.0

This document is intended for hardware and software engineers integrating the CM4/5 TPM Module into security-critical embedded systems.

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1 Overview

The **CM4/5 TPM Module** is a compact hardware security stack-up designed specifically for the Raspberry Pi® Compute Module 4 (CM4) and Compute Module 5 (CM5). It integrates the Infineon OPTIGA™ TPM SLB 9670VQ2.0, a TPM 2.0-compliant Trusted Platform Module, directly onto a carrier PCB that mechanically and electrically interfaces with the CM4/5 high-density connectors.

The board provides a hardware root of trust for Linux-based embedded security applications including secure boot, remote attestation, disk encryption (e.g. via TPM-backed LUKS), platform configuration measurement, and cryptographic key storage.

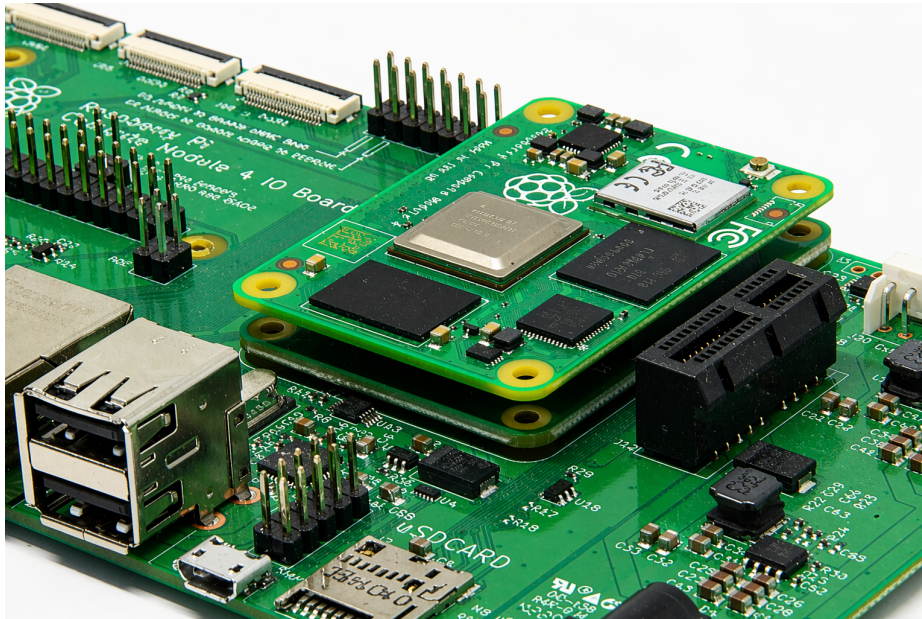


Figure 1: CM4/5 TPM adapter board stacked-up with Raspberry Pi® compute module

1.1 Key Features

- Infineon OPTIGA™ TPM SLB 9670VQ2.0 — TCG TPM 2.0 compliant
- SPI interface, up to 43 MHz clock rate at 3.3 V
- FIPS 140-2 Level 2 validated (Certificate #3492)
- CC EAL4+ certified hardware security core
- Compatible with Raspberry Pi CM4 and CM5 via dual high-density board-to-board connectors
- 3.3 V single-supply operation with internal low-power management
- Built-in Linux kernel support (no proprietary drivers required)
- 24 PCRs (SHA-1 and SHA-256)
- Hardware Random Number Generator (NIST SP800-90A)
- Full Endorsement Key (EK) personalization with EK certificate
- Standard operating temperature: -20 to $+85^{\circ}\text{C}$
- Compact stackable form factor; four M2.5 mounting holes

1.2 Applications

- Secure boot and measured boot (e.g. with UEFI or U-Boot TPM support)
- Full-disk encryption with TPM-backed key unsealing
- Remote attestation and device identity
- IoT edge security gateways
- Industrial embedded computing requiring hardware RoT
- Government and defense embedded platforms

2 Ordering Information

Table 1: Device Variants

Part Number	Description	Temp. Range	Notes
CM45-TPM-R10	CM4/5 TPM Module Rev. 1.0	-20 to +85 °C	Standard version, SLB 9670VQ2.0

3 Connector and Signal Description

3.1 Board-to-Board Connectors

The CM4/5 TPM Module uses 4 100-pin high-density board-to-board connectors (J1, J2, J3, J4) that are compatible with the standard Raspberry Pi CM4/CM5 connector footprint. These mate directly with the corresponding connectors on the Compute Module.

Table 2: Board Connectors

Ref.	Type	Pitch / Pins	Part number
J1 & J4	High-density B2B	0.4 mm / 100	DF40C-100DS-0.4V(51)
J2 & J3	High-density B2B	0.4 mm / 100	DF40C-100DP-0.4V(51)

3.2 TPM Signals Mapped from CM4/5

The following signals are routed from the CM4/5 SPI bus and GPIO through the board-to-board connectors to the SLB 9670 TPM IC. The receptacle connector pin numbers (as labelled in the schematic) are listed alongside the TPM IC pin numbers.

Table 3: TPM Signal Mapping

TPM IC Pin	Signal	Receptacle Pin	Direction	Description
19	SCLK	38	Input	SPI clock (SPI mode 0 only)
20	$\overline{\text{CS0}}$	39	Input	Chip select 0, active low; default $\overline{\text{CS}}$
20	$\overline{\text{CS1}}$	37	Input	Chip select 1, active low; not populated (DNP)
21	MOSI	44	Input	SPI data from CM4/5 to TPM
24	MISO	40	Output	SPI data from TPM to CM4/5
17	RST#	45	Input	Reset, active low; connect to CM4/5 GPIO24
18	PIRQ#	41	Output	Interrupt request, active low, open-drain

SPI mode 0 (CPOL=0, CPHA=0) is the only mode supported by the SLB9670. The CM4/5 SPI controller must be configured accordingly in the device tree overlay.

4 Electrical Characteristics

4.1 Absolute Maximum Ratings

Exposure to conditions beyond the absolute maximum ratings may cause permanent device damage. These are not operating conditions.

Table 4: Absolute Maximum Ratings

Parameter	Min	Typ	Max	Unit	Condition
Supply voltage V_{DD}	-0.3	3.3	5.0	V	—
Voltage on any pin	-0.3	—	$V_{DD}+0.3$	V	—
Ambient temperature	-20	—	+85	°C	Standard device
Storage temperature	-40	—	+125	°C	—
ESD (HBM, 1.5 k Ω / 100 pF)	—	—	2000	V	EIA/JESD22-A114-B
ESD (CDM)	—	—	500	V	STM5.3.1-1999
Latch-up immunity	—	—	100	mA	EIA/JESD78

4.2 Functional Operating Range

Table 5: Functional Operating Range

Parameter	Min	Typ	Max	Unit	Condition
Supply voltage V_{DD}	3.0	3.3	3.6	V	—
Ambient temperature	-20	—	+85	°C	Standard device
Useful / operating lifetime	—	—	10	yr	Avg. $T_A = 55^\circ\text{C}$

4.3 Current Consumption

Table 6: TPM IC Current Consumption ($T_A = 25^\circ\text{C}$, $V_{DD} = 3.3\text{ V}$)

Parameter	Min	Typ	Max	Unit	Condition
Active mode current I_{VDD_Active}	—	—	25	mA	During command execution
Sleep mode current I_{VDD_Sleep}	—	110	—	μA	CS# inactive; no SPI transaction

The SLB 9670 automatically enters a low-power sleep state after each completed command/response transaction. No explicit sleep command is required.

4.4 SPI Interface DC Characteristics

Table 7: DC Characteristics — SPI Pins (SCLK, CS#, MISO, MOSI, RST#, PIRQ#), $V_{DD} = 3.3\text{ V}$

Parameter	Min	Typ	Max	Unit	Condition
Input high voltage V_{IH}	$0.7 V_{DD}$	—	$V_{DD}+0.5$	V	SCLK, MISO, MOSI, CS#
Input low voltage V_{IL}	-0.5	—	$0.3 V_{DD}$	V	SCLK, MISO, MOSI, CS#
Input leakage current	-150	—	+150	μA	SCLK, CS#, MISO, MOSI
Output high voltage V_{OH}	$0.9 V_{DD}$	—	—	V	$I_{OH} = -100\ \mu\text{A}$
Output low voltage V_{OL}	—	—	$0.1 V_{DD}$	V	$I_{OL} = 1.5\text{ mA}$
Pad input capacitance C_{IN}	—	—	10	pF	—
Output load capacitance C_{LOAD}	—	—	40	pF	MISO

4.5 SPI Interface AC Characteristics

Table 8: AC Characteristics — SPI Interface, $V_{DD} = 3.3\text{ V}$

Parameter	Symbol	Min	Max	Unit	Condition
SCLK frequency	f_{CLK}	—	43	MHz	$t_{SLEW} \geq 1\text{ V/ns}$
SCLK frequency	f_{CLK}	—	38	MHz	$t_{SLEW} < 1\text{ V/ns}$
CS# high time	t_{CS}	50	—	ns	—
CS# setup time	t_{CSS}	5	—	ns	CS# fall to SCLK rise
CS# hold time	t_{CSH}	5	—	ns	SCLK fall to CS# rise
MOSI setup time	t_{SU}	2	—	ns	To SCLK rising edge
MOSI hold time	t_H	3	—	ns	From SCLK rising edge
MISO valid delay	t_V	0	$0.7 t_{CLKL}$	ns	From SCLK falling edge

5 TPM Properties

Table 9: TPM 2.0 Capabilities (SLB 9670VQ2.0)

Property	Value
TPM specification	TCG TPM 2.0 (Family “2.0”)
Manufacturer ID (TPM_PT_MANUFACTURER)	"IFX"
Platform Configuration Registers (PCRs)	24 (SHA-1 and SHA-256)
Free NV memory	$\geq 6,962$ bytes
Loaded sessions (min)	3 (TPM_PT_HR_LOADED_MIN)
Active sessions (max)	64 (TPM_PT_ACTIVE_SESSIONS_MAX)
Loaded transient objects (min)	3 (TPM_PT_HR_TRANSIENT_MIN)
Loaded persistent objects (min)	7 (TPM_PT_HR_PERSISTENT_MIN)
NV counters	≤ 8
Command/response buffer	$\leq 1,024$ bytes
NV read/write size	≤ 768 bytes
I/O buffer	1,420 bytes
RNG standard	NIST SP800-90A
FIPS validation	FIPS 140-2 Level 2 (Certificate #3492)
CC certification	EAL4+
Linux kernel support	Built-in (no out-of-tree driver needed)

6 Reset Timing

Table 10: Device Reset Timing

Parameter	Symbol	Min	Max	Unit
Cold (power-on) reset delay	t_{POR}	—	80	μs
Warm reset pulse width	t_{WRST}	2	—	μs
Reset inactive time (before first command)	t_{RSTIN}	60	—	ms

Important: RST# must not be asserted within the t_{RSTIN} window after deassertion. Violating this timing during power-on or warm boot may trigger the TPM's dictionary-attack protection mechanisms and potentially lock the device. Always issue TPM2.Shutdown before asserting RST#. TPM commands must only be started after t_{RSTIN} has expired.

7 Mechanical Dimensions

Table 11: Board Mechanical Specification

Parameter	Value	Notes
PCB outline	55 mm × 40 mm	
Layer count	2	Standard FR4 stackup
PCB thickness	1.6 mm	Standard
Mounting holes	4 × M2.5	3.5 mm from each board edge
Corner radius	4 × R3.5 mm	
Connector J1 / J4	Centered at 25 mm from left edge	Top board edges
Connector J2 / J3	Centered at 25 mm from left edge	Bottom board edges
Surface finish	ENIG	RoHS compliant
Solder mask colour	Green	—

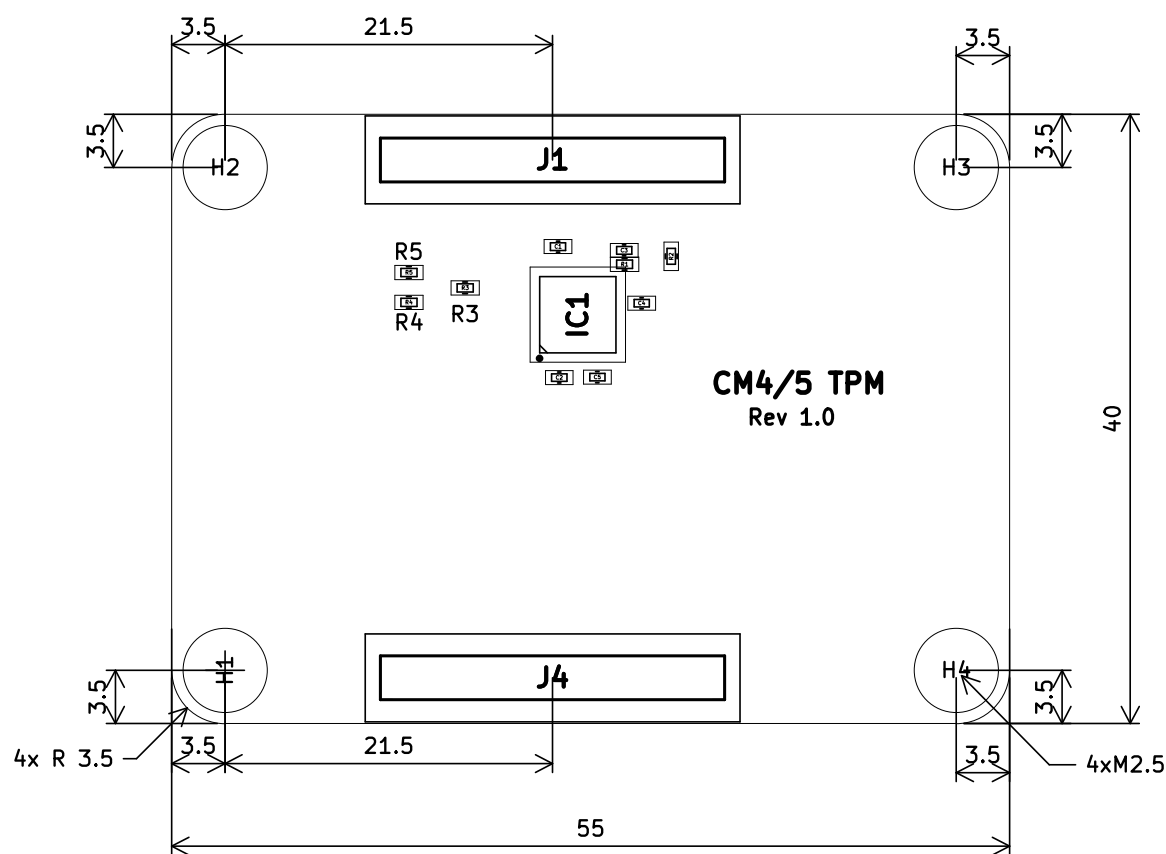


Figure 2: CM4/5 TPM Module Rev. 1.0 — PCB dimension drawing (all dimensions in mm). Board outline: 55 × 40 mm. Mounting holes: 4 × M2.5, 3.5 mm from edges. Corner radii: 4 × R3.5 mm.

8 Software Integration

8.1 Linux Device Tree Overlay

The TPM is connected via SPI on the Compute Module. A device tree overlay must be enabled to expose the TPM to the Linux TPM stack. For Raspberry Pi OS with CM4/5, add the following to `/boot/firmware/config.txt`:

```
dtparam=spi=on
dtoverlay=tpm-slb9670
```

8.2 Verifying TPM Detection

After boot, verify the TPM is enumerated:

```
ls /dev/tpm*
# Expected: /dev/tpm0 /dev/tpmrm0

dmesg | grep -i tpm
# Should show: tpm_tis_spi spi0.1: 2.0 TPM (device-id 0x1B, rev-id 22)

tpm2_getcap properties-fixed | grep -E "MANUFACT|VENDOR|FIRMWARE"
```

8.3 Required Packages

```
sudo apt install tpm2-tools libtss2-dev
```

8.4 Secure Boot / LUKS Integration

For TPM-backed full-disk encryption with LUKS2:

```
# Enroll LUKS key to TPM2 (systemd-cryptenroll, Debian/Ubuntu)
sudo systemd-cryptenroll --tpm2-device=auto --tpm2-pcrs=0+7 /dev/sdX

# Or with clevis/tang for network-bound disk encryption
sudo clevis luks bind -d /dev/sdX tpm2 '{"pcr_ids":"7"}'
```

For more software information, please refer to github repo <https://github.com/kahlenberg/CM5-TPM>

9 Compliance and Certifications

Table 12: Standards Compliance

Standard / Certification	Details
TCG TPM 2.0	Family “2.0”, compliant to TCG Spec Rev. 01.16 and Rev. 01.38
FIPS 140-2 Level 2	Certificate #3492 (16 July 2019)
Common Criteria	EAL4+ certified hardware security core
RoHS	PCB and components are RoHS compliant
Microsoft Windows	TPM qualifies for Windows platform certification
Google Chromebook	Meets Chromebook TPM certification criteria
Intel TXT	Compatible with Intel Trusted Execution Technology

10 References

- [1] Infineon Technologies AG, *OPTIGA™ TPM SLB 9670 TPM2.0 Data Sheet*, Rev. 1.5, 2024-08-07.
- [2] TCG, *Trusted Platform Module Library (Parts 1–4)*, Family 2.0, Level 00, Rev. 01.38, 2016-09-29.
- [3] TCG, *TCG PC Client Platform TPM Profile (PTP) Specification*, Family 2.0, Level 00, Rev. 01.03 v22, May 2017.
- [4] Raspberry Pi Ltd., *Compute Module 4 Datasheet*.
- [5] Raspberry Pi Ltd., *Compute Module 5 Datasheet*.

Revision History

Rev.	Date	Author	Changes
1.0	September 12, 2026	—	Initial release

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